

# Business Problem Statement

Project Title: Customer Churn Analysis

## Client Brief

A subscription-based service provider is facing challenges with customer retention. While the company has steady new sign-ups, management lacks clarity on churn drivers, retention patterns, and revenue at risk. An analytics engagement has been initiated to uncover insights that can reduce churn, improve retention, and optimize revenue.

## Problem Statement

The current business operations face challenges in:

Customer attrition is impacting recurring revenue, but the business currently has no unified, trustworthy view of who is churning, why, and at what cost. The raw datasets exhibit data-quality issues --- that prevent reliable measurement. In addition, churn behavior, plan value, and support-escalation history live in separate tables with no shared reporting layer, making it difficult to answer basic questions such as which plan types churn the most, how much monthly revenue is at risk, and whether customer complaints and escalations are correlated with churn.

This has resulted in missed opportunities for proactive retention strategies and revenue optimization.

## Objectives

- Analyze churn rate across plans, states, and customer segments.
- Identify retention opportunities and escalation patterns.
- Optimize customer engagement strategy to reduce churn.
- Provide actionable insights for revenue protection.

## Key Performance Indicators (KPIs)

### 1. Churn Rate Analysis:

Calculate the churn rate ( $\text{Churned Customers} \div \text{Total Customers}$ ).

### 2. Churn by Plan Type Analysis:

Calculate the churn rate for each plan type — Basic, Standard, and Premium.

### 3. Churn by State Analysis:

Calculate the churn rate grouped by country and state.

### 4. Retention Rate Analysis:

Calculate the retention rate ( $1 - \text{Churn Rate}$ )

### 5. ARPU (Average Revenue Per User) Analysis:

Calculate ARPU —  $\text{SUM}(\text{monthly\_charges}) \div \text{COUNT}(\text{active\_customerid})$

### 6. Average Customer Tenure Analysis:

Calculate average customer tenure —  $\text{AVG}(\text{DATEDIFF}(\text{cancellation\_date OR NOW()}, \text{subscription\_start\_date}))$

## 7. Revenue at Risk Analysis:

Calculate revenue at risk —  $\text{SUM}(\text{monthly\_charges}) \text{ WHERE churn\_score} > 70$ .

## 8. Escalation Rate Analysis:

Calculate the escalation rate —  $(\text{SUM}(\text{escalations}) \div \text{COUNT}(\text{complaints})) \times 100$ .

## 9. Average Complaints per Customer Analysis:

Calculate average complaints per customer —  $\text{COUNT}(\text{complaints}) \div \text{COUNT}(\text{DISTINCT customerid})$

## 10. Escalations → Churn Correlation Analysis:

Calculate churn rate for customers with escalations  $\geq 1$  versus customers with escalations = 0

# Data Preparation & Transformation

## Data Cleaning

- The analyst should check the customer, subscription, and support tables for missing values, duplicates, and inconsistent entries.
- Dates, names, and categories (like gender, plan type, or state) should be put into one clear format.
- Any fixes or assumptions made during cleaning should be written down so the process is transparent.

## Feature Engineering

- **Age** — derived from date of birth (DOB) in the customer table.
- **Churn Flag** — derived as Churned / Active, based on presence of a cancellation date.
- **Tenure (months)** — subscription start date to cancellation date (or current date, if active).
- **Complaint Count** — number of support complaints raised per customer.
- **Escalation Flag** — whether a customer's complaint was escalated.
- **CSAT Band** — bucketed CSAT score (e.g., Low / Medium / High satisfaction).
- **Revenue at Risk** — monthly charges attributable to currently active, high churn-score customers.

## Aggregations

- Plan-level churn performance
- State-level churn performance
- Time-based churn trends

## Dashboard Requirements

- The Power BI report has 6 pages with a navigation panel on every page, slicers and a Reset button on the Home page.
- 1. **Home:** KPI cards (Total Customers, Churn Rate, Escalation Rate, MRR, Revenue at High Risk, Revenue Lost); Key Review Points and Business Recommendations; slicers for Plan Type, Subscription Type, Gender and Country.

- **2. Customer Segmentation:** Total, churned, active and active High-Risk customers; risk band distribution; churn by age group and gender; total customers by plan type.
- **3. Churn Overview:** Total churned customers and churn rate; cancellation reason breakdown; churn rate by plan type; yearly churn trend.
- **4. Risk & Complaints:** Total and escalated complaints; churn rate by risk band and by complaint count; % High Risk by country; risk band by plan type; slicers for Country and Contract Type.
- **5. Revenue Impact:** ARPU, Average CLTV, Revenue at High Risk and Revenue Lost; revenue lost by cancellation reason, risk band and year; slicers for Customer Status and Contract Type.
- **6. Time Trends:** Churned customers and revenue lost by cancellation year; revenue at risk by subscription year and risk band; slicers for Cancellation Reason and Contract Type.

## **Deliverables**

### **Dashboard**

An interactive Power BI dashboard that tells the churn story clearly. It should have filters (slicers) and drill-through options so users can explore churn by plan, location, or customer segment.

### **Insights Summary**

A short report that highlights the main findings — what is driving churn, where growth or retention opportunities exist, and what operational changes could help.